C12.7 DVM Operations

This section describes main DVM Operations:

- *TLB Invalidate*
- *Branch Predictor Invalidate* on page C12-290
- *Physical Instruction Cache Invalidate* on page C12-291
- *Virtual Instruction Cache Invalidate* on page C12-292
- *Synchronization* on page C12-293
- *Hint* on page C12-293.

C12.7.1 TLB Invalidate

This section lists the TLB Invalidate operations that the DVM message supports.

Table C12-5 shows the fixed values for the TLB Invalidate message fields.

Table C12-5 TLB Invalidate message fixed values

ARADDR bit	Value	Status
[(n-1):32][a]	SBZ	Reserved
[15]	`0b0`	Completion not required
[7]	SBZ	Reserved
[4:1]	SBZ	Reserved

a. n is the width of the AR address bus.

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Table C12-6 shows the TLB Invalidate message, **ARADDR[14:12]** = 0b000 and the encoding for the supported operations. See *DVM transactions format* on page C12-286 for further information on the message encoding.

Table C12-6 Supported TLB Invalidate operations

ARADDR bit						Operation
[14:12] Message type	[11:10] Hypervisor	[9:8] Security	[6] VMID	[5] ASID	[0] VA	
0b000	0b10 All Guest OS	0b10 Secure	0b0 Ignore	0b0 Ignore	0b0 Ignore	Secure TLB Invalidate all
			0b0 Ignore	0b0 Ignore	0b1 Match	Secure TLB Invalidate by VA
			0b0 Ignore	0b1 Match	0b0 Ignore	Secure TLB Invalidate by ASID
			0b0 Ignore	0b1 Match	0b1 Match	Secure TLB Invalidate by ASID and VA
	0b10 All Guest OS	0b11 Non-secure	0b0 Ignore	0b0 Ignore	0b0 Ignore	All OS TLB Invalidate all
			0b1 Match	0b0 Ignore	0b0 Ignore	Guest OS TLB Invalidate all
			0b1 Match	0b0 Ignore	0b1 Match	Guest OS TLB Invalidate by VA
			0b1 Match	0b1 Match	0b0 Ignore	Guest OS TLB Invalidate by ASID
			0b1 Match	0b1 Match	0b1 Match	Guest OS TLB Invalidate by ASID and VA
	0b11 Hypervisor	0b11 Non-secure	0b0 Ignore	0b0 Ignore	0b0 Ignore	Hypervisor TLB Invalidate all
			0b0 Ignore	0b0 Ignore	0b1 Match	Hypervisor TLB Invalidate by VA

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C12.7.2 Branch Predictor Invalidate

This section lists the Branch Predictor Invalidate operations that the DVM message supports.

Table C12-7 shows the fixed values for the Branch Predictor Invalidate message fields.

Table C12-7 Branch Predictor Invalidate message fixed values

ARADDR bit	Value	Status
[(n-1):32][a]	SBZ	Reserved
[15]	0b0	Completion not required
[11:10]	0b00	Applies to all Guest OS and Hypervisor
[9:8]	0b00	Applies to Secure and Non-secure
[7]	SBZ	Reserved
[6]	0b0	VMID is specified on **ARADDR[31:24]**
[5]	0b0	ASID is specified on **ARADDR[23:16]**
[4:1]	SBZ	Reserved

a. n is the width of the AR address bus.

Table C12-8 shows the Branch Predictor Invalidate message, **ARADDR[14:12]** = 0b001 and the encodings for the supported operations. See *DVM transactions format* on page C12-286 for further information on the message encoding.

Table C12-8 Supported Branch Predictor Invalidate operations

ARADDR bit		Operation
[14:12] Message type	[0] VA	
0b001	0b0 Ignore	Branch Predictor Invalidate all
	0b1 Match	Branch Predictor Invalidate by VA

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C12.7.3 Physical Instruction Cache Invalidate

This section lists the Physical Instruction Cache Invalidate operations that the DVM message supports.

Table C12-9 shows the fixed values for the Physical Instruction Cache Invalidate message fields.

Table C12-9 Physical Instruction Cache Invalidate message fixed values

ARADDR bit	Value	Status
[(n-1):32][a]	SBZ	Reserved
[15]	0b0	Completion not required
[11:10]	0b00	Applies to all Guest OS and Hypervisor
[7]	SBZ	Reserved
[4:1]	SBZ	Reserved

a. n is the width of the AR address bus

Table C12-10 shows the Physical Instruction Cache Invalidate message, **ARADDR[14:12]** = 0b010 and the encodings for the supported operations. See *DVM transactions format* on page C12-286 for further information on the message encoding.

Table C12-10 Supported Physical Instruction Cache Invalidate operations

ARADDR bit				Operation
[14:12] Message type	[9:8] Security	[6:5][a] Virtual Index	[0] VA	
0b010	0b10 Secure	0b00	0b0 Ignore	Secure Physical Instruction Cache Invalidate all
		0b00	0b1 Match	Secure Physical Instruction Cache Invalidate by PA without Virtual Index
		0b11	0b1 Match	Secure Physical Instruction Cache Invalidate by PA with Virtual Index
	0b11 Non-secure	0b00	0b0 Ignore	Non-secure Physical Instruction Cache Invalidate all
		0b00	0b1 Match	Non-secure Physical Instruction Cache Invalidate by PA without Virtual Index
		0b11	0b1 Match	Non-secure Physical Instruction Cache Invalidate by PA with Virtual Index

a. If **ARADDR[6]** is 0b1 then Virtual Index VA[27:20] at **ARADDR[31:24]** is used as part of the physical address
If **ARADDR[5]** is 0b1 then Virtual Index VA[19:12] at **ARADDR[23:16]** is used as part of the physical address

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C12.7.4 Virtual Instruction Cache Invalidate

This section lists the Virtual Instruction Cache Invalidate operations that the DVM message supports.

Table C12-11 shows the fixed values for the Virtual Instruction Cache Invalidate message fields.

Table C12-11 Virtual Instruction Cache Invalidate message fixed values

ARADDR bit	Value	Status
(n-1):32[a]	SBZ	Reserved
[15]	0b0	Completion not required
[7]	SBZ	Reserved
[4:1]	SBZ	Reserved

a. n is the width of the AR address bus

Table C12-12 shows the Virtual Instruction Cache Invalidate message, **ARADDR[14:12]** == 0b011 and the encodings for the supported operations. See *DVM transactions format* on page C12-286 for further information on the message encoding.

Table C12-12 Supported Virtual Instruction Cache Invalidate operations

ARADDR bit						Operation
[14:12] Message type	[11:10] Hypervisor	[9:8] Security	[6] VMID	[5] ASID	[0] VA	
0b011	0b00 Hypervisor and All Guest OS	0b00 Secure and Non-secure	0b0 Ignore	0b0 Ignore	0b0 Ignore	Invalidate all. Applies to Secure and Non-secure. Applies to Hypervisor and all Guest OS.
		0b11 Non-secure	0b0 Ignore	0b0 Ignore	0b0 Ignore	Invalidate all. Applies to Non-secure. Applies to Hypervisor and all Guest OS.
	0b10 All Guest OS	0b10 Secure	0b0 Ignore	0b1 Match	0b1 Match	Secure Invalidate by ASID and VA.
		0b11 Non-secure	0b1 Match	0b0 Ignore	0b0 Ignore	Guest OS, Invalidate all.
				0b1 Match	0b1 Match	Guest OS, Invalidate by ASID and VA.
	0b11 Hypervisor	0b11 Non-secure	0b0 Ignore	0b0 Ignore	0b1 Match	Hypervisor, Invalidate by VA.

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C12.7.5 Synchronization

This section lists the Synch Operation that the DVM message supports.

Table C12-13 shows the fixed values for the Sync message fields.

Table C12-13 Sync message fixed values

ARADDR bit	Value	Status
[(n-1):32][a]	SBZ	Reserved
[15]	0b1	Completion Required
[11:10]	0b00	Applies to all Guest OS and Hypervisor
[9:8]	0b00	Applies to Secure and Non-secure
[7]	SBZ	Reserved
[6]	0b0	Ignore VMID
[5]	0b0	Ignore ASID
[4:1]	SBZ	Reserved
[0]	0b0	Virtual address is not specified in this message

a. n is the width of the AR address bus

Note

The Sync message is the only supported message type that has the Completion Required field, **ARADDR[15]** set to 1.

Table C12-14 shows the message type encoding for the Sync Operation and usage cases.

Table C12-14 Supported Sync Operations

ARADDR bit [14:12] Message type	Operation
0b100	Synchronization

C12.7.6 Hint

A reserved message address space is provided for future Hint messages.

Table C12-15 shows the fixed values for the Hint message fields.

Table C12-15 Hint message fixed values

ARADDR bit	Value	Status
[15]	0b0	Completion not required

All Hint messages must respond with the snoop response value **CRRESP** set to 0 on the CR channel.

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Table C12-16 shows the message type encoding for future Hint operations.

Table C12-16 Support for future Hint operations

ARADDR bit [14:12] Message type	Operation
0b110	Reserved

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Chapter C13
Interface Control

This chapter describes the optional signals that can be used to configure the behavior of the ACE interface. It contains the following section:

- *About the interface control signals* on page C13-296.

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